

# Taste shaped the use of botanical drugs

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## Abstract

The perception of taste and flavour (a combination of taste, smell and chemesthesis) here also referred to as chemosensation, enables animals to find high-value foods and avoid toxins. Humans have learned to use unpalatable and toxic substances as medicines, yet the importance of chemosensation in this process is poorly understood. Here we generate tasting-panel data for botanical drugs and apply phylogenetic generalised linear mixed models to test whether intensity and complexity of chemosensory qualities as well as particular tastes and flavours can predict ancient Graeco-Roman drug use. We found chemosensation to be strongly predictive of therapeutic use: botanical drugs with high therapeutic versatility have simple yet intense tastes and flavours, and 21 of 22 chemosensory qualities predicted at least one therapeutic use. In addition to the common notion of bitter tasting medicines, we also found starchy, musky, sweet, cooling, and soapy drugs associated with versatility. In ancient Greece and Rome, illness was thought to arise from imbalance in bodily fluids or humours, yet our study suggests that uses of drugs were based on observed physiological effects that are often consistent with modern understanding of chemesthesis and taste receptor pharmacology.

### eLife assessment

This **valuable** study links the "taste" of botanicals to their application as medicines used by the ancient Greco-Roman society. The authors used phylogenetic linear mixed models in a Bayesian framework to test the relationships between taste qualities, intensities, complexities, and therapeutic use. The evidence supporting the conclusions is **solid**, although there is a minor weakness concerning the somewhat inconsistent method of botanical preparation and presentation to the taster panelists; subjective bias and robustness of the participants' responses might have been overlooked. The study may be of broad interest to pharmacologists and scientists working on drug discovery, particularly those interested in natural products.

## Introduction

‘A bitter pill to swallow’ is just one of many common idioms referring to the taste of medicine, which from ancient to early modern times has relied mostly on botanical drugs (Touwaide, 2022). In ancient Greece and Rome, the expected effects of botanical drugs were explained by their taste (Jones, 1959 [↗](#); Einarson and Link, 1990 [↗](#); Jouanna and Allies, 2012) and according to the prevailing medical theory, disease resulted from a disequilibrium between bodily fluids or humours including phlegm, blood, yellow bile and black bile; drugs affected the flow and balance of these humours (Jones, 1959 [↗](#); King, 2013). Theophrastus (300 BCE) attributed sweet the capacity to smoothen, astringent with the power to desiccate and solidify, pungent with the capacity to cut or to separate out the heat, salty with desiccating and irritating properties, and bitter with the capacity to melt and irritate (Einarson and Link, 1990 [↗](#)). Other systems of medicine, such as Ayurveda and Traditional Chinese Medicine continue to classify *materia medica* according to tastes and flavours (Dragos and Gilca, 2018 [↗](#); Shou-zhong, 2018), and indigenous societies use chemosensory qualities to select botanical drugs (Casagrande, 2000 [↗](#); Leonti et al., 2002 [↗](#); Shepard, 2004 [↗](#)).

The perception of taste and flavour (a combination of taste, smell and chemesthesis) here also referred to as chemosensation, has evolved to meet nutritional requirements and are particularly important in omnivores for seeking out nutrients and avoiding toxins (Rozin and Todd, 2016 [↗](#); Breslin, 2013 [↗](#); Glendinning, 2022 [↗](#)). The rejection of bitter stimuli has generally been associated with the avoidance of toxins (Glendinning, 1994 [↗](#); Lindemann, 2001 [↗](#); Breslin, 2013 [↗](#)) but to date no clear relationship between bitter compounds and toxicity at a nutritionally relevant dose could be established (Glendinning, 1994 [↗](#); Nissim et al., 2017 [↗](#)). While bitter tasting metabolites occurring in fruits and vegetables have been linked with a lower risk for contracting cancer and cardiovascular diseases (Drewnoswski and Gomez-Carneros, 2000), the avoidance of pharmacologically active compounds is probably the reason why many medicines, including botanical drugs, taste bitter (Johns, 1990 [↗](#); Mennella et al., 2013 [↗](#)).

Botanical drugs are chemically complex and often used for a range of different health problems (Wagner et al., 2007 [↗](#)). Chemical complexity determines pharmacological complexity, since different chemicals interact with different targets (Gertsch, 2011 [↗](#)), and influence taste complexity and intensity (Spence and Wang, 2018 [↗](#); Breslin and Beauchamp, 1997 [↗](#); Green et al., 2010). Today, some associations between chemosensory qualities and therapeutic uses, such as bitter tasting botanical drugs for stomach problems or astringent ones for diarrhoea, are still recognised in Western herbal medicine (Wagner et al., 2007 [↗](#)). These observations establish a role for chemosensory qualities in medicine. Yet, taste and flavour are largely irrelevant to modern medicine, aside from the pharmaceutical industry looking for ways to mask bad tastes (Mennella et al., 2013 [↗](#)).

Whether ancient systems of traditional medicine could be relevant to modern science is questionable. The theory of humours was the principal underlying concept of Western medicine from ancient Graeco-Roman times until the Industrial Age, but as an entirely discredited theory it might be expected that associated practices also lack scientific basis. However, we propose that taste and flavour perception of botanical drugs links humoral theory to modern medicine. We hypothesize that chemosensation and underlying physiological effects mediated by chemesthesis and taste receptor pharmacology co-explain the diversity of therapeutic uses of botanical drugs in ancient times. We further hypothesize that associations between perceived complexity and intensity of chemosensory qualities of botanical drugs and their versatility (number of categories of therapeutic use a botanical drug is used for) provide insights into the development of empirical pharmacological knowledge and therapeutic behaviour. If tastes and flavours, as determined by a modern-day tasting panel, do predict uses as recorded in ancient texts, this would be an unprecedented insight into how chemosensation was translated into physiological and therapeutic expectations and how it guided human behaviour in pre-scientific times. To test these hypotheses,

we assessed the relationships between chemosensory qualities, their intensity and complexity with therapeutic uses of botanical drugs described in Pedanios Dioscorides' pharmacognostic compendium, *De Materia Medica* (*DMM*; Matthioli, 1967-1970; Staub et al., 2016), the foremost pharmaceutical source of antiquity. *DMM* compiles knowledge on the sourcing, preparation and therapeutic uses of botanical drugs traded and used in the Eastern Mediterranean region of the Roman Empire during the 1st century CE (Riddle, 1985). We collected 700 botanical drugs described in *DMM*, (**Supplementary file 1**) and used quality descriptors to represent chemosensation by a trained tasting panel scoring each botanical drug for presence and intensity of each quality (**Figure 1**; **Supplementary files 2 and 3**). We arranged the botanical drugs into therapeutic uses according to affected organs, therapeutic functions, diseases, and symptoms as indicated in *DMM*, and into broader and more inclusive categories of therapeutic use (**Figure 1**; **Supplementary files 1 and 4**). By applying phylogenetic generalised linear mixed models (PGLMMs) to our data and a plant phylogeny (Zanne et al., 2014; see Methods) we tested whether perceived taste and flavour qualities, as well as their overall intensity and complexity predict recorded therapeutic uses whilst simultaneously accounting for the confounding effects of the shared uses and chemosensory stimuli by drugs from closely related species.

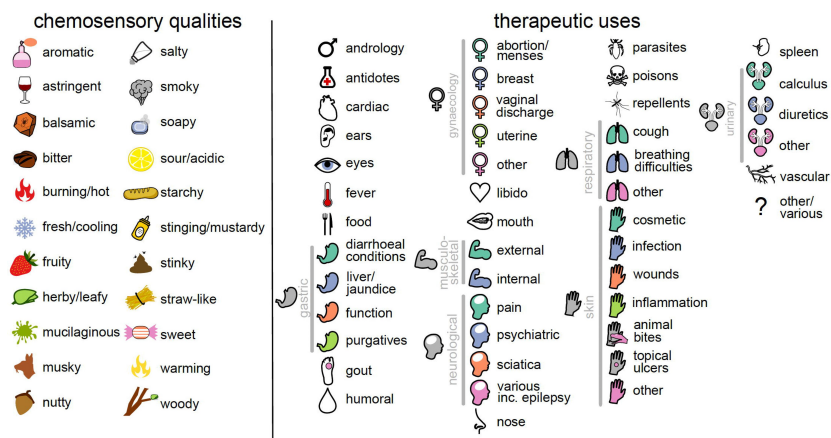
## Results

### Diversity of chemosensory qualities and therapeutic uses

We used 22 descriptors to represent perceived chemosensory qualities of 700 botanical drugs and identified 46 therapeutic uses across 25 categories as described in *DMM* (**Figure 1**; **Supplementary files 1 and 4**). A group of eleven panellists conducted 3973 sensory trials (see Methods) and reported 10,463 individual perceptions of qualities. The most frequently reported qualities were bitter (1556 reports; 39% of all assessed samples), herby/leafy (1146 reports; 29%), aromatic (795 reports; 20%) and stinging (730 reports; 18%), while the least was mucilaginous (128 reports; 3.2%). Weak qualities were most frequently reported (5244 reports; 50%); 3899 (37%) of reports were of medium strength and 1320 (13%) were perceived as strong. Balsamic (69 reports; 24%), burning/hot (41 reports; 22%) and fresh/cooling (49 reports; 21%) were the qualities most often perceived as strong, while smoky (87 reports; 61%), sour (224 reports; 60%) and starchy (190 reports; 59%) the qualities most often perceived as weak. Panellists perceived between zero (e.g., *Anemone coronaria*, root) and ten (*Cinnamomum verum*, bark) individual qualities per drug. The mean number of therapeutic uses per botanical drug was 6.3 and the most frequently mentioned therapeutic use being 'gynaecology – abortion and menses' (272 use records) and the least frequently mentioned 'cardiac problems' (seven use records).

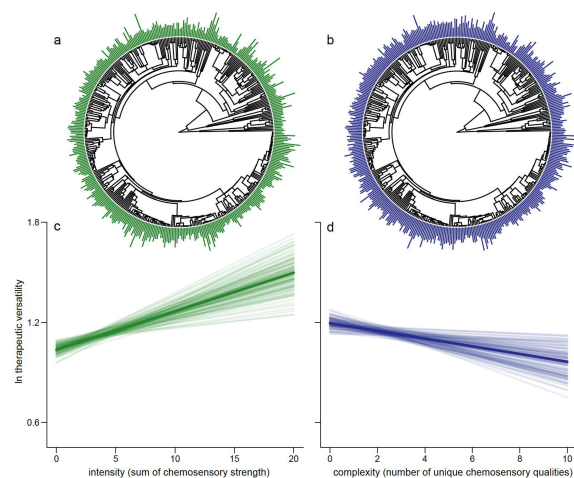
### Simple and intense chemosensory perception is associated with therapeutic versatility

The relationship between intensity of chemosensation (the sum of all chemosensory scores across all qualities, **Figure 2A**; see Methods) and therapeutic versatility is positive ( $p_x < 0.001$ ; **Figure 2C**). On the other hand, chemosensory complexity (the number of taste and flavour qualities a botanical drug is scored for, **Figure 2B**; see Methods) is negatively associated with how many different out of the 25 categories of therapeutic use botanical drugs are used for (e.g., cardiac, gynaecological, humoral, henceforth referred to as 'therapeutic versatility' ( $p_x = 0.026$ ; **Figure 2D**). Importantly, the multiple regression framework we use allows for each trait to covary whilst simultaneously predicting their underlying relationship with therapeutic versatility.



**Figure 1.**

The 22 chemosensory qualities and 46 therapeutic uses studied here. Each chemosensory quality and use is represented by an icon that is used throughout the manuscript. Therapeutic uses that share an icon are considered to represent the same category of use (25 in total); these are linked by grey bars.



**Figure 2.**

Drugs eliciting strong overall chemosensory perception (intense drugs) associated with relatively few perceived unique tastes (less complex) predict the number of categories of therapeutic use (therapeutic versatility). We show the distribution of intensity (**A**) and complexity (**B**) at the tips of the phylogenetic tree. As some species are represented by multiple parts, an average is shown here for representative purposes; each plant part is considered separately in the analysis. We plot the mean regression line as estimated from the parameter estimates of our Bayesian phylogenetic regression analyses (dark line) along with 100 random samples of the posterior distribution (faded lines). For graphical representation, the slopes are estimated holding other effects at their mean. We do this for both the significant positive association between intensity and therapeutic versatility (**C**,  $p_x < 0.001$ ) and the significant negative association between chemosensory complexity and therapeutic complexity (**D**,  $p_x = 0.008$ ).

The online version of this article includes the following figure supplement for **Figure 2**. **Figure supplement 1.** The relationship between individual therapeutic uses and chemosensory intensity/complexity.

## Strength of specific chemosensory qualities predict therapeutic versatility

All models we analyse have strong phylogenetic signal (modal  $h^2 > 0.9$ , see Methods) which may have led to misleading impressions about medicinal tastes and flavours in the past in analyses which did not account for such statistical non-independence. Indeed, while we do find a significant positive association between the strength of bitterness and therapeutic versatility ( $p_x = 0.013$ ), the most versatile drugs were not only the bitter-tasting ones; contrary to popular concepts of bad-tasting medicine (Mennella et al., 2013). We also find that the perceived strengths of starchy ( $p_x = 0.017$ ), musky ( $p_x = 0.013$ ), sweet ( $p_x = <0.018$ ), and soapy ( $p_x = 0.002$ ) qualities were positively associated with therapeutic versatility while the intensity of perceived sourness was negatively associated with versatility ( $p_x = <0.001$ ).

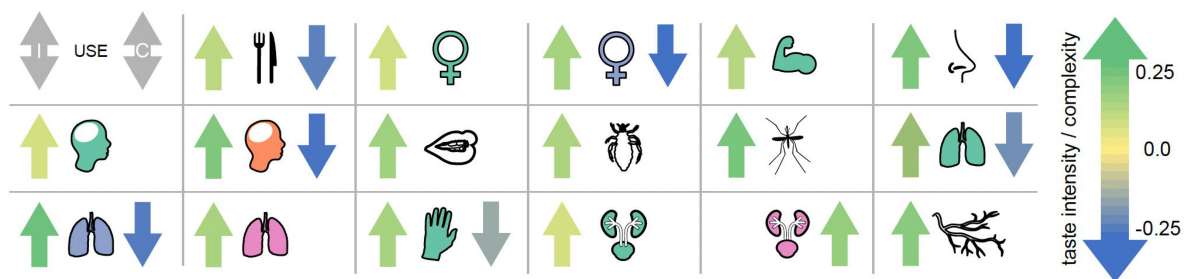
## Associations between specific chemosensory qualities and specific uses

All therapeutic uses (with the exception of ‘gynaecology – other’) are significantly associated with at least one specific chemosensory quality, whether positively (99 instances) or negatively (50 instances) and often with high magnitude of effect (Figure 3; Supplementary file 5). Thus, the perceived presence or absence of several specific qualities significantly confer the chemosensory perception of medicine (less than 5% of their estimated parameter distribution crosses zero,  $p_x < 0.05$ ) with high magnitude of effect (Figure 3).

## Discussion

Unexpectedly, botanical drugs eliciting fewer but intense chemosensations were more versatile (Figure 2). People often associate complexity with intensity, and taste complexity is popularly interpreted with a higher complexity of ingredients (Spence and Wang, 2018). However, chemosensory perception is relative to exposed stimuli and simple tastes can be associated with complex chemistry when intense tastes mask weaker tastes, or when tastants are blended (Breslin and Beauchamp, 1997; Green et al., 2010). For example, starchy flavours or sweet tastes can be sensed when bitter and astringent antifeedant compounds are present below a certain threshold while salts enhance overall flavour by suppressing the perception of bitter tastants (Breslin and Beauchamp, 1997; Johns, 1990). On the other hand, combinations of different tastants or olfactory stimuli do not necessarily result in increased perceived complexity (Spence and Wang, 2018; Weiss et al., 2012). The relationship between intense chemosensation and versatility (Figure 2) is consistent with a perceived causal link between strong sensations and strong effects that is probably innate (Ganchrow et al., 1983; Glendinning, 1994). We also detected nuances in significance, and complete absence of significance across the relationships between individual therapeutic uses and complexity/intensity magnitudes for which we lack, however, more specific explanations (Figure 2–Figure supplement 1).

Low therapeutic versatility of sour tasting drugs (12 negative associations) may best be understood by considering the uses that this taste was negatively associated with in our PGLMMs (Figure 3). For example, the significant association between sour and treatment of coughs ( $p_x = 0.038$ ) and other respiratory problems ( $p_x = 0.042$ ; Figure 3; Supplementary files 5 and 6) might be related to coughing triggered by acid signalling in the airways (Kollarik et al., 2007). Moreover, acidic solutions and tissue acidosis are associated with inflammation and nociception while acid reflux is a common cause of chronic cough probably triggered by an oesophageal-bronchial reflex (Irwin, 2006). The negative associations of sour but also salty drugs with pain management ( $p_x = 0.021$  and  $0.022$ , respectively; Figure 3; Supplementary files 5 and 6), are probably related to the avoidance of nociception known by personal experience to result from the contact between acidic or salty solutions and skin or mucous membrane lesions.



**Figure 3.**

The magnitude of effect for 22 different specific chemosensory qualities across 46 different therapeutic uses. **(A)** A heat map showing the strength of the association between specific qualities (columns) and therapeutic uses (rows). Where we find a significant association between the strength of chemosensory quality and whether a drug is used for a particular therapeutic purpose in our phylogenetic binary (probit) response model, the corresponding cell is coloured according to its estimated parameter value; see colour scale in panel **(B)**. All other variables are zeroed and coloured white for visualization. **(B)** An overall 'magnitude of effect' is calculated for each chemosensory quality by summing the significant betas ( $p_x < 0.05$ , i.e., those displayed in the heat-map above) across all different uses and plotted as bars coloured by their magnitude; see colour scale alongside axis. The sum of all betas regardless of significance are shown as grey bars. Note that the large parameter estimate for "smoky" is owing to the fact that no botanical drug with any smoky quality (scored by any participant) is ever used for androgynous therapeutic purposes and thus is poorly estimated in that model (**Supplementary files 1,2,5 and 6**).



The therapeutic versatility of drugs perceived as sweet (13 positive associations) and starchy (12 positive associations) (**Figure 3**; **Supplementary files 5 and 6**) is influenced by the many staples and edible fruits that are also used for medicine (e.g., protein-rich Fabaceae seeds were often perceived as having starchy tastes). The daily use of food items, generally with a low toxicity, has allowed for a more detailed comprehension of their postprandial effects, which arguably has led to a diversification of uses. The 96 botanical drugs specifically mentioned also for food (though, there are more than 150 edible drugs in our dataset; **Supplementary file 1**) show positive associations with starchy ( $p_x = 0.005$ ), nutty ( $p_x = 0.002$ ) and salty ( $p_x = 0.001$ ) and negative associations with bitter ( $p_x = 0.007$ ), woody ( $p_x = 0.001$ ) and stinging ( $p_x = 0.033$ ) chemosensory qualities.

Sweet sensations are known to induce analgesia (Kakeda et al., 2010) reflected here by a positive association of sweet tasting drugs with the treatment of pain ( $p_x = 0.032$ ). Both starchy ( $p_x = 0.001$ ) and salty ( $p_x = 0.002$ ) are associated with the treatment of diarrhoea and dysentery. Today, mixtures of glucose, starch and electrolytes are used to treat secretolytic and inflammatory diarrhoea caused by bacterial enterotoxins (Thiagarajah et al., 2015; Binder et al., 2014). These oral rehydration solutions exploit sodium/glucose co-transport by sodium/glucose co-transporter 1 in the small intestine and sodium absorption through sodium–hydrogen antiporter 2 stimulated by short chain fatty acids synthesized from starch by bacteria in the colon (Thiagarajah et al., 2015; Binder et al., 2014). Drugs perceived as starchy are also positively associated with the treatment of topical ulcers ( $p_x = 0.001$ ), skin infections ( $p_x = 0.025$ ), other skin conditions ( $p_x = 0.035$ ) and the topical treatment of venomous and non-venomous animal bites ( $p_x = 0.001$ ) where plasters with lenitive properties might bring benefit. Moreover, fresh Fabaceae seeds contain protease inhibitors (Birk, 1996; Wink and Waterman, 1999) such as trypsin and chymotrypsin inhibitors, which have the potential to inhibit bacterial proteases and control bacterial virulence during cutaneous infections (Frees et al., 2013; Culp and Wright, 2017).

Bitter was the most frequently reported chemosensory quality and showed nine significant associations with therapeutic use – including the negative association with foods (**Figure 3**). Though many bitter compounds are toxic, not all bitter plant metabolites are (Glendinning, 1994; Drewnoswski and Gomez-Carneros, 2000; e.g., iridoids, flavonoids, glucosinolates, bitter sugars). In part, this may be the outcome of an arms race between plant defence and herbivorous mammals' bitter taste receptor sensitivities, resulting in the synthesis of metabolites capable of repelling herbivores and confounding the perception of potential nutrients by mimicking tastes of toxins. Here, poisons showed no association with bitter but positive associations with aromatic ( $p_x = 0.041$ ), sweet ( $p_x = 0.022$ ) and soapy ( $p_x = 0.025$ ) as well as a negative association with salty ( $p_x = 0.046$ ) qualities.

Anti-inflammatory pharmacology of several bitter tasting compounds explains the positive associations with treatments of sciatica ( $p_x = <0.001$ ; **Figure 3**). Bitter tasting drugs to treat sciatica derive from several disparate lineages containing a range of different anti-inflammatory compound classes such as iridoids (Viljoen et al., 2012; *Ajuga, Centaurium*), sesquiterpene lactones (Coricello, 2020; *Artemisia, Inula*), triterpene saponines (Gepdiremen et al., 2006; Wang et al., 2014; *Citrullus, Ecballium, Leontice*), and coumarins (Kirsch et al., 2016; *Opopanax, Ruta*) (**Supplementary files 1 and 2**).

Generally, overarching explanations for versatility based on chemosensory perception are not feasible. However, we hypothesize that many of the associations we found (**Figure 3**; **Supplementary files 5 and 6**) are grounded in observed physiological effects mediated by chemesthesis and taste receptor pharmacology (Chandrashekar et al., 2006; Palmer and Servant, 2022). Cumulative evidence suggests that together with transient receptor potential channels (TRPs), extraoral taste receptors and ectopic olfactory receptors form a chemosensory network involved in homeostasis, disease response and off-target drug effects (Foster et al., 2014; An and Liggett, 2018; Kang and Koo, 2012; Hauser et al., 2017; Clark et al., 2012). Bitter taste

receptors, for instance, are located also in human airways where their activation mediates bronchodilation and the movement of motile cilia which leads to improved respiration and evacuation of mucus (Shah et al., 2009 [DOI](#); Deshpande et al., 2010 [DOI](#)). Indeed, bitter showed positive associations with the treatment of breathing difficulties ( $p_x = 0.027$ ) and other respiratory problems ( $p_x = 0.004$ ). We find that drugs used to treat pain, including pain associated with breastfeeding (breast inflammation and lactation), are significantly more likely to have fresh and cooling qualities ( $p_x = 0.01$  and  $0.037$ , respectively). This is likely owing to interactions with TRP channels (McKemy et al., 2002 [DOI](#); Behrendt et al., 2004 [DOI](#); Proudfoot et al., 2006 [DOI](#); Memon et al., 2019 [DOI](#)). Specifically, the TRPM8 channel mediates cold stimuli and is known to be activated by cooling essential oil constituents producing analgesic effects (McKemy et al., 2002 [DOI](#); Behrendt et al., 2004 [DOI](#); Proudfoot et al., 2006 [DOI](#)). Cooling menthol and eucalyptol have been shown to act as counterirritants inhibiting respiratory irritation caused by smoke constituents (Willis et al., 2011 [DOI](#)) explaining the positive association between drugs used to treat breathing difficulties with fresh and cooling flavours ( $p_x = 0.04$ ). The strong positive association of burning and hot drugs with oral applications for treating musculoskeletal problems ( $p_x = 0.004$ ) might be related to analgesic properties mediated by desensitizing effects upon TRP channel interaction (Marwaha et al., 2016 [DOI](#); Aroke et al., 2020 [DOI](#)). Burning and hot drugs are also positively associated with vascular problems ( $p_x = 0.042$ ) including varices and haemorrhoids and can be explained with vasorelaxant effects transmitted by TRP channels on the endothelial cells (Montell, 2005 [DOI](#); Pires and Earley, 2017 [DOI](#)). TRPV channels are also involved in kidney and urinary bladder functioning (Montell, 2005 [DOI](#)) and TRPV1-knockout mice urinate more frequently without voiding their bladder (Birder et al., 2002 [DOI](#)). Interactions with TRPV1 and other TRPV receptors might therefore explain the use of hot botanical drugs to treat other urinary problems including urinary retention, incontinence, kidney and bladder problems ( $p_x = 0.047$ ).

The concepts of humoral medicine may have derived from the observation that the appearance and consistency of bodily fluids during disease change and that they can be modified. The positive association of bitter taste with humoral management ( $p_x = 0.004$ ) as well as with liver problems and jaundice ( $p_x = 0.002$ ) is explained by uses intended to purge and resolve bitter bile and seems to include what we call choleric and cholekinetics today, such as the roots of the great yellow gentian (*Gentiana lutea*) or the herb of the common vervain (*Verbena officinalis*). Astringency and woody qualities showed positive associations with the treatment of diarrhoea and dysentery ( $p_x = 0.029$  and  $p_x = 0.019$ ) where there is a need for controlling the flow of fluids and for drying them up. Similarly woody, often correlating with astringency (Scalbert et al., 1989 [DOI](#)) was also positively associated with the staunching of vaginal discharge ( $p_x = <0.001$ ). Curiously, astringency showed positive associations with external applications for musculoskeletal conditions ( $p_x = 0.001$ ) where drying internal bleeds and solidifying tissues and bones are needed. Negative associations of astringency emerged with conditions where the flux of humours, perspiration and the evacuation of liquids and mucus are treatment strategies, such as in fever ( $p_x = 0.045$ ), humoral management ( $p_x = 0.02$ ), urinary calculus ( $p_x = 0.02$ ), other urinary problems (retention, incontinence, kidney and bladder problems,  $p_x = <0.001$ ) and cough ( $p_x = 0.002$ ). More obviously, the intake of salt aggravates fluid retention and oedema, and this is likely the reason why remedies for dropsy and oedema (diuretics) are negatively associated with salty tastes ( $p_x = 0.023$ ). Chemosensory qualities of botanical drugs and their physiological effects probably explain the persistence of humoral medicine until the 19th century.

Looking forwards, our study highlights the potential of chemosensory pharmacology in the development of herbal medicines. Botanical drugs with observed chemosensory profiles not aligned with those of their therapeutic use might represent drugs working beyond chemosensory pharmacology. Conversely, associations of chemosensory qualities with therapeutic uses of botanical drugs may certainly overlap with the presence of metabolites with specific therapy-relevant constituents acting beyond chemosensory pharmacology. This is, for instance, the case with opium (dried milk sap of *Papaver somniferum* L.) indicated for ‘cough’ and perceived as bitter (Figure 3 [DOI](#); Supplementary files 1 and 2) but at the same time containing alkaloids with central



cough suppressant properties (Dewick, 2002 [↗](#)). Besides the associations discussed, other therapeutically relevant chemosensory receptor functions might be involved in the associations currently lacking explanations. While we are aware that the limitation of this study is the impossibility to address all associations found and the dependence on published pharmacological data for the explanatory part, our approach demonstrates the power of phylogenetic methods applied to human behaviour and assessment of plant properties. We show that chemosensory perception connects ancient therapeutic knowledge with modern science, even though aetiologies were fundamentally different in Graeco-Roman times. Our study offers a blueprint for macroscale re-evaluation of pre-scientific knowledge and points towards a central role for taste and flavour in medicine.

## Methods

### Botanical fieldwork

A total of 700 botanical drugs (ca. 70% of the botanical drugs described in *DMM*) associated with 407 species were collected from the wild, cultivated in home gardens or purchased from commercial sources between 2014 and 2016. Plant collection included at least one voucher per species and one or more bulk samples of the therapeutic plant part (botanical drug). Latin binomials follow *plantsoftheworldonline.org* [↗](#). Botanical vouchers were deposited at the herbarium of the University of Geneva (G) and the herbarium of the National and Kapodistrian University of Athens (ATHU). Bulk samples for sensory analysis were dried at 40–60° C and stored in plastic containers. Collection permits were obtained from the Greek Ministry of Environment and Energy (6Ω8K4653Π8-AK7).

### Data extraction from *DMM*

Information about the therapeutic uses (use records) of botanical drugs were extracted from Matthioli's 1568 edition of Dioscorides' *De Materia Medica* (Matthioli, 1967-1970; Staub et al., 2016). Botanical drugs were arranged into 46 therapeutic uses defined by organ, therapeutic function, disease and symptom across 25 broader categories of use (**Figure 1** [↗](#); **Supplementary files 1 and 4**). Each of the 700 botanical drugs was then assigned a binary variable defining whether it was recommended (1) or not (0) for each of the 46 therapeutic uses based on their descriptions in *DMM*. When the description in *DMM* permitted identification (Taxon ID; **Supplementary file 1**) only to genus level (due to ambiguities or the possibility that different closely related taxa were subsumed) we chose one species as the representative for the whole genus (**Supplementary file 1**: 'Taxon panel'). Unidentified taxa or taxa identified only to the family level were not considered in this analysis. The category 'food' includes only those drugs where edibility was specifically mentioned in *DMM* and by far not all those that are used as food. As a measure of therapeutic versatility, we summed up the total number of broader categories of therapeutic use each botanical drug was recommended for.

### Tasting panel

The chemosensory qualities of each of the 700 botanical drugs were assessed experimentally by trained human subjects using the sensory analysis technique Conventional Profiling (ISO 11035) performed in accordance with the Declaration of Helsinki (World Medical Association of Helsinki, 2013) and the ethics guidelines of the Institute of Food Science & Technology (*Institute of Food Science and Technology*, 2020 [↗](#)). The study design was approved by the Ethics Committee of the University Hospital of Cagliari (NP/2016/4522). Informed consent was obtained from all research participants. We used 22 chemosensory descriptors to represent taste and flavour perception (**Figure 1** [↗](#); **Supplementary files 2 and 3**). The evaluation took place at the Hospital of *San Giovanni di Dio* (Cagliari, Italy) during a period of five months in 2016. The panel consisted of eleven healthy Caucasian volunteers (six males and five females; age 30.7±7.4). Working language

was Italian. Based on a random subset of 40 (out of 700) samples, four initial training sessions were held to establish a consensus regarding chemosensory terminology. Hedonic (e.g., “good” or “bad”), self-referential (e.g., “minty” for mint) and infrequent descriptors were excluded to reduce subjectivity and to increase discriminatory power of the analysis. Synonymous and closely related descriptors (e.g., “astringent” and “tannic”) were pooled to reduce attribute redundancy.

Samples (0.1 to 2 g of dried drug pieces) were labelled randomly (using a random number generator) with three-digit codes, assigned randomly and double-blinded (in case of common drugs impossible) to individual panellists. Not all drugs were tasted by all panellists while random distribution permitted that individual panellists were challenged with repeated samples. Samples were presented in identical 125 ml plastic containers at room temperature. The relatively big range in quantity of dispensed samples owes to toxicological concerns and practical reasons but in any case, were tailored to permit the perception of their chemosensory qualities. Panellists were instructed to chew the amount of sample necessary for chemosensory perception, to annotate perceived qualities and intensities and to spit out residues of samples, and finally rinse their mouth with drinking water. The breaks between tasting different samples depended on the persistence of chemosensory perception. For each sensory trial a separate evaluation sheet was used. Quality intensities were evaluated on a 4- point ordinal scale: absent (0), weak (1), medium (2) and strong (3). Overall, 3973 individual sensory trials were conducted, with an average of  $361 \pm 153$  trials per panellist and  $5.7 \pm 1.3$  trials per botanical drug. On average each panellist tasted 17,2 drugs per hour using 10.5 sessions (18 sessions in total) lasting approximately two hours each. From the ordinal data, two additional metrics were calculated to represent chemosensory *complexity* (the total number of different chemosensory qualities a botanical drug was assigned to as non-zero by each panellist) and chemosensory *intensity* (the sum of all chemosensory scores assigned across all qualities by each panellist).

## Phylogenetic tree

We used the species-level supertree of land plants to represent the relationships among the taxa extracted from *DMM* (Zanne et al., 2014 [link](#)). Some species were not found in the tree; in these cases, species were attached to the most recent common ancestor of the respective genus or, if absent, the family. Branch lengths of added taxa were set to retain ultrametricity. Tree manipulations were made in R (R Core Team, 2016 [link](#)) using the packages APE and PHYTOOLS (Paradis et al., 2004 [link](#); Revell, 2012 [link](#)). The trees are represented in **Figure 2** [link](#) alongside the chemosensory intensity and complexity data.

## Statistical procedure

In order, to determine whether chemosensation and therapeutic uses were linked, we used two sets of phylogenetic linear mixed models implemented in a Bayesian framework within the R package MCMCglmm (Hadfield, 2010 [link](#)). The first set of models considered therapeutic versatility (number of categories of therapeutic use) modelled as a zero-truncated poisson distributed response variable. We ran two models: (i) considering all 22 individual chemosensory qualities as ordinal-scale independent variables and (ii) considering chemosensory intensity and complexity as independent variables. Together, these models tell us whether any particular chemosensory quality as well as whether intense or complex chemosensation predict how often a drug is used for any therapeutic purpose. We then ran a second set of probit models which considered each individual therapeutic use as a binary response. That is, we ran 46 models examining the relationship between individual chemosensory qualities and therapeutic uses (one for each use, where all qualities are included as independent variables) and an additional 46 to test the relationship between each therapeutic use and chemosensory intensity and complexity.

The significance of all parameters is assessed using the proportion of the posterior distribution of parameter estimates that cross zero ( $p_x$ ). If a parameter has a substantial effect on a model, we expect the distribution of parameter estimates to be shifted away from zero (i.e.,  $p_x < 0.5$ ). All

models were run for a total of 500,000 iterations after burn-in, sampling every 5,000.

Phylogenetic signal was calculated using heritability ( $h^2$ ) which has an identical interpretation to Pagel's lambda. All models showed high heritability values ( $h^2 > 0.9$ ) demonstrating the importance of species' shared ancestry ( $h^2$  ranges between 0 and 1, and values close to 1 indicate very strong phylogenetic signal). In all models we included plant part, panellist ID, and the phylogenetic matrix as random effects. The inclusion of plant part and the phylogenetic matrix allowed for the fact that in some cases, multiple drugs were extracted from different parts of the same plant (e.g., *Vitis vinifera* L., fruit; leaf, herb, seeds) whilst still accounting for the overall underlying evolutionary relationships. The inclusion of panellist ID removed the effect of any biased perception by single panellists across drugs as well as any differences among panellists for an individual drug. We also included the number of days between sample collection and when each sample was tasted, in order, to account for any degradation of samples over time that may have affected chemosensory perception. For all fixed factors in all models, we used largely uninformative priors (implementing a normal distribution with a mean of 0 and a variance of  $1e10$ ). Residual variance was estimated using an inverse gamma prior ( $V = 1$ ,  $\nu = 0.002$ ). We used parameter-expanded priors (roughly uniform standard deviations) for all random effects including phylogenetic variance (Hadfield and Nakagawa, 2010 [↗](#)).

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## Author Contributions

Marco Leonti, Conceptualization; Methodology, Formal analysis; Investigation; Resources; Data curation; Writing – original draft preparation; Writing – review & editing; Supervision; Project administration; Funding acquisition; Joanna Baker, Methodology; Software; Validation; Formal analysis; Investigation; Writing – original draft preparation; Writing – review & editing; Visualisation; Peter Staub, Conceptualization; Methodology; Investigation; Data curation; Laura Casu, Methodology; Resources; Data curation; Project administration; Julie Hawkins, Investigation; Writing – original draft preparation; Writing – review & editing; Supervision.

## Competing Interest Statement

The authors declare no competing interest.

## Supplementary files

Supplementary file 1. Sampled botanical drugs and their uses *ex* Matthioli (1568).

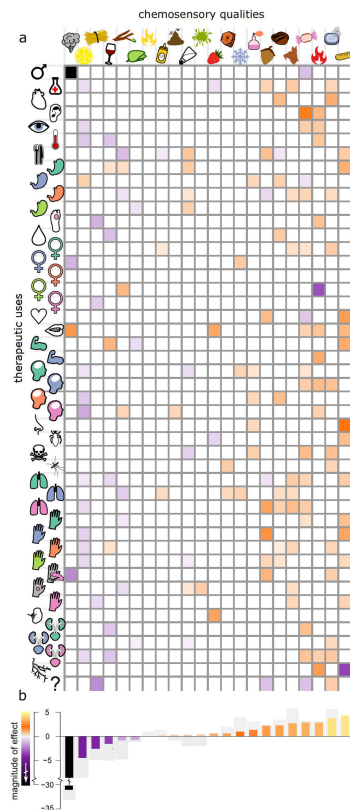
Supplementary file 2. Sampled botanical drugs and their chemosensory qualities (panel data).

Supplementary file 3. Codes and the short descriptions of the 22 chemosensory descriptors to represent tastes and flavours.

Supplementary file 4. Codes and the short descriptions of the therapeutic uses.

Supplementary file 5. Heat map showing the strength of the association between chemosensory quality and therapeutic use.

Supplementary file 6. Significance of individual parameter estimates from the models of individual chemosensory quality against individual therapeutic use calculated as the proportion of the parameter estimate crossing zero,  $p_x$ .



**Figure 2–figure supplement 1.**

The relationship between individual therapeutic uses and chemosensory intensity/complexity. Chemosensory intensity and complexity were significantly associated with seventeen uses. No therapeutic use was predicted by weaker qualities. An arrow is shown indicating the direction of any significant association ( $p_x < 0.05$ ) with either chemosensory intensity (left) or chemosensory complexity (right). An arrow points upward if a drug is more likely to be used for a given purpose with increasing chemosensory intensity or complexity. Arrows are shaded by the strength of the relationship (magnitude of estimated parameter, see scale). For seven individual uses more intense and less complex chemosensation increased the probability of a drug being used i.e., there was a preference for stronger and simpler qualities. Nine uses had a significant positive association with chemosensory intensity and were predicted by stronger qualities alone. A single use (various urinary problems) was uniquely significantly associated with increased chemosensory complexity. No therapeutic use was predicted by weaker qualities.

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**Editors**

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**Reviewer #1 (Public Review):**

Summary: The authors explored correlations between taste features of botanical drugs used in ancient times and therapeutic uses, finding some potentially interesting associations between intensity and complexity of flavors and therapeutic potential, plus some more specific associations described in the discussion section. I believe the results could be of potential benefit for the drug discovery community, especially for those scientists working in the field of natural products.

Strengths:

Owing to its eclectic and somehow heterodox nature, I believe the article might be of interest for a general audience. In fact, I have enjoyed reading it and my curiosity was raised by the extensive discussion.

The idea of revisiting a classical vademecum with new scientific perspectives is quite stimulating.

The authors have undertaken a significant amount of work, collecting 700 botanical drugs and exploring their taste and association with known uses via eleven trained panellists.

Weaknesses:

I have some methodological concerns. Robustness in the panelists' perceptions has not been addressed, and not every panellist tasted every drug because of time constraints. The breaks between tasting different samples was not standardized, and depended on the persistence of chemosensory perception, possibly also due to time constraints.

- <https://doi.org/10.7554/eLife.90070.2.sa1>

#### **Reviewer #2 (Public Review):**

##### **Summary:**

This is an unusual, but interesting approach to link the "taste" of plants and plant extracts to their therapeutic use in ancient Graeco-Roman culture. The authors used a panel of 11 trained tasters to test ~700 different medicinal plants and describe them in terms of 22 "taste" descriptors. They correlated these descriptors with the plant's medical use as reported in the De Materia Medica (DMM 1st Century, CE). Correcting for some of the plants' evolutionary phylogenetic relationships, the authors found that taste descriptors along with intensity measures were correlated with the "versatility" and/or a specific therapeutic use of the medicine. For example, simple but intense tastes were correlated with versatility of a medicine. Specific intense tastes were linked to versatility while others were not; intense bitter, starchy, musky, sweet, cooling and soapy were associated with versatility, but sour and woody were negatively associated. Also some specific tastes could be associated with specific uses - both positive and negative associations. Some of these findings make sense immediately, but others are somewhat surprising, and the authors propose some links between taste and medicinal use (both historical and modern use) in the discussion. The authors state that this study allows for a re-evaluation of pre-scientific knowledge, pointing toward a central role for taste in medicine.

##### **Strengths:**

The real strength of this study is the novelty of this approach - using modern day tasters to evaluate ancient medicinal plants to understand the potential relationships between taste and therapeutic use, lending some support to the idea that the "taste" of a medicine is linked to its effectiveness as a treatment.

##### **Weaknesses:**

Because of the limitations of time and the type of botanicals being tested, there is an inherent difficulty in assessing taste intensity. However, because these botanicals are tested by multiple panelists and sometimes tested repeatedly by individual panelists, this helps support the author's analyses.

- <https://doi.org/10.7554/eLife.90070.2.sa0>

#### **Author Response**

The following is the authors' response to the original reviews.

##### **Public Reviews:**

##### **Reviewer #1 (Public Review):**

##### **Summary:**

*The authors explored correlations between taste features of botanical drugs used in ancient times and therapeutic uses, finding some potentially interesting associations between intensity and complexity of flavors and therapeutic potential, plus some more specific associations described in the discussion sections. I believe the results could be of*



*potential benefit to the drug discovery community, especially for those scientists working in the field of natural products.*

*Strengths:*

*Owing to its eclectic and somehow heterodox nature, I believe the article might be of interest to a general audience. In fact, I have enjoyed reading it and my curiosity was raised by the extensive discussion.*

*The idea of revisiting a classical vademecum with new scientific perspectives is quite stimulating.*

*The authors have undertaken a significant amount of work, collecting 700 botanical drugs and exploring their taste and association with known uses via eleven trained panelists.*

*Weaknesses:*

*I have some methodological concerns. Was subjective bias within the panel of participants explored or minimized in any manner?*

Yes, in all models we included ‘panellist’ as a random effect and therefore any biased perception by a single panellist across drugs or differences among panellists for an individual drug was accounted for. We now make this clearer in our methods.

*Were the panelists exposed to the drugs blindly and on several occasions to assess the robustness of their perceptions?*

The study was double blind, but blinding was not possible with the more well-known drugs (e.g., almonds, walnuts, thyme, mint). A random number generator was used to assign the drugs to the panellists, and according to the random distribution, some drugs were presented to the same panellist more than once. Robustness of panellists’ perception was not assessed specifically. We have added some text to the methods to clarify.

*Judging from the total number of taste assessments recorded and from Supplementary Material, it seems that not every panelist tasted every drug. Why?*

Because there were many drugs and panellists had time constraints. Overall, 3973 individual sensory trials were conducted, with an average of  $361 \pm 153$  trials per panellist and  $5.7 \pm 1.3$  trials per botanical drug.

*It may be a good idea to explore the similarity in the assessments of the same botanical drug by different volunteers. If a given descriptor was reported by a single volunteer, was it used anyway for the statistical analysis or filtered out?*

All responses were used as reported by the panellists, including potential ‘outliers’. As described above, the inclusion of ‘panellist’ as a random effect means that if one individual gave an unusual description of a particular drug in comparison to other individuals, this would be less impactful on any parameter estimates.

*The idea of "versatility" is repeatedly used in the manuscript, but the authors do not clearly define what they call "versatile".*

In line with suggestions made by reviewers, we have slightly adjusted the definition of therapeutic versatility and have now clearly defined the term on first use. Here, we define therapeutic versatility as the number of therapeutic ‘categories’ a drug is used for (the 25

broad categories are represented by shared iconography in Figure 1). Our revised results include analyses using this definition – which are qualitatively identical to our previous results which defined versatility using the 46 individual therapeutic uses.

*The introduction should be expanded. There are plenty of studies and articles out there exploring the evolution of bitter taste receptors, and associating it with a hypothetical evolutionary advantage since bitter plants are more likely to be poisonous.*

We agree. Bitter is arguably the most frequent chemosensory attribute of plants and botanical drugs perceived by humans. Our data shows that ‘poisons’ are not associated with bitterness but positively with ‘aromatic’, ‘sweet’ and ‘soapy’ – and negatively with ‘salty’ qualities.

We have added this paragraph to the introduction:

"The perception of taste and flavour (a combination of taste, smell and chemesthesis) here also referred to as chemosensation, has evolved to meet nutritional requirements and are particularly important in omnivores for seeking out nutrients and avoiding toxins (Rozin and Todd, 2016; Breslin, 2013; Glendinning, 2022). The rejection of bitter stimuli has generally been associated with the avoidance of toxins (Glendinning, 1994; Lindemann, 2001; Breslin, 2013) but to date no clear relationship between bitter compounds and toxicity at a nutritionally relevant dose could be established (Glendinning, 1994; Nissim et al., 2017). While bitter tasting metabolites occurring in fruits and vegetables have been linked with a lower risk for contracting cancer and cardiovascular diseases (Drewnoswski and Gomez-Carneros, 2000) the avoidance of pharmacologically active compounds is probably the reason why many medicines, including botanical drugs, taste bitter (Johns, 1990; Mennella et al., 2013)."

And expanded in the discussion:

"Though many bitter compounds are toxic, not all bitter plant metabolites are (Glendinning, 1994; Drewnoswski and Gomez-Carneros, 2000; e.g., iridoids, flavonoids, glucosinolates, bitter sugars). In part, this may be the outcome of an arms race between plant defence and herbivorous mammals' bitter taste receptor sensitivities, resulting in the synthesis of metabolites capable of repelling herbivores and confounding the perception of potential nutrients by mimicking tastes of toxins. Here, poisons showed no association with bitter but positive associations with aromatic (px = 0.041), sweet (px = 0.022) and soapy (px = 0.025) as well as a negative association with salty (px = 0.046) qualities."

*Since plant secondary metabolites are one of the most important sources of therapeutic drugs and one of their main functions is to protect plants from environmental dangers (e.g., animals), this evolutionary interplay should be at least briefly discussed in the introductory section.*

This is now referred to in the introduction as well as in the discussion.

*Since the authors visit some classical authors, Paracelsus' famous quote "All things are poison and nothing is without poison. Solely the dose determines that a thing is not a poison" may be relevant here. Also note that some authors have explored the relationship between taste receptors and pharmacological targets (e.g., Bioorg Med Chem Lett. 2012 Jun 15;22(12):4072-4).*

We agree that pharmacologic action is determined by the dose. We now refer to the dose in the introduction: "...to date no clear relationship between bitter compounds and toxicity at a nutritionally relevant dose could be established (Glendinning, 1994; Nissim et al., 2017)".

We are aware of the fact that several authors have explored the relationship between taste receptors as targets and their similarity with other targets. We use many examples from the literature to explain our data. Our analysis did, however, not highlight any association between sweet tastes and epilepsy (as reported in *Bioorg Med Chem Lett.* 2012 Jun 15;22(12):4072-4)). We are not able to explain all associations, and we acknowledge that there may be more associations between chemosensory receptors and therapeutic effects than those found and discussed here.

**Reviewer #2 (Public Review):**

*Summary:*

*This is an unusual, but interesting approach to link the "taste" of plants and plant extracts to their therapeutic use in ancient Graeco-Roman culture. The authors used a panel of 11 trained tasters to test ~700 different medicinal plants and describe them in terms of 22 "taste" descriptors. They correlated these descriptors with the plant's medical use as reported in the De Materia Medica (DMM 1st Century, CE). Correcting for some of the plants' evolutionary phylogenetic relationships, the authors found that taste descriptors along with intensity measures were correlated with the "versatility" and/or specific therapeutic use of the medicine. For example, simple but intense tastes were correlated with the versatility of a medicine. Specific intense tastes were linked to versatility while others were not; intense bitter, starchy, musky, sweet, cooling, and soapy were associated with versatility, but sour and woody were negatively associated. Also, some specific tastes could be associated with specific uses - both positive and negative associations. Some of these findings make sense immediately, but others are somewhat surprising, and the authors propose some links between taste and medicinal use (both historical and modern use) in the discussion. The authors state that this study allows for a re-evaluation of pre-scientific knowledge, pointing toward a central role of taste in medicine.*

*Strengths:*

*The real strength of this study is the novelty of this approach - using modern-day tasters to evaluate ancient medicinal plants to understand the potential relationships between taste and therapeutic use, lending some support to the idea that the "taste" of a medicine is linked to its effectiveness as a treatment.*

*Weaknesses:*

*While I find this study very interesting and potentially insightful into the development and classification of certain botanical drugs for specific medicinal use, I would encourage the authors to revise the manuscript and the accompanying figures significantly to improve the reader's understanding of the methods, analyses, and findings. A more thorough discussion of the limitations of this particular study and this general type of approach would also be very important to include.*

Figures were revised, one deleted (former Fig. 3), and another one put to the supplementary (former Fig. 4, now Figure supplement 1). We now acknowledge limitations in the final paragraph.

*The metric of versatility seems somewhat arbitrary. It is not well explained why versatility is important and/or its relationship with taste complexity or intensity.*

We have modified the definition of versatility in line with reviewers' comments. We have provided a detailed explanation of this in our response to reviewer #1 but for ease of

reference, we paste this again here:

Here, we define therapeutic versatility as the number of therapeutic ‘categories’ a drug is used for (the 25 broad categories are represented by shared iconography in Figure 1). Our revised results include analyses using this definition – which are qualitatively identical to our previous results which defined versatility using the 46 individual therapeutic uses.

The importance of versatility was not the focus but the impact of taste intensity and complexity on versatility. We hypothesize that associations between perceived complexity and intensity of chemosensory qualities with versatility of botanical drug use provides insights into the development of empirical pharmacological knowledge and therapeutic behaviour (now included in the introduction).

*Similarly, the rationale for examining the relationships between individual therapeutic uses and taste intensity/complexity is not well explained, and given that a similar high intensity/low complexity relationship is common for most of the therapeutic uses, it restates the same concepts that were covered by the initial versatility comparison.*

The examination of the relationships between individual therapeutic uses and taste intensity/complexity fine-tunes the overall analysis and shows that this concept is applicable in general. However, in general, the reviewer is correct, and this is not our main focus. We therefore shifted the analysis including the figure to the supplementary material and state in the discussion: “We also detected nuances in significance, and complete absence of significance across the relationships between individual therapeutic uses and complexity/intensity magnitudes for which we lack, however, more specific explanations (Figure supplement 1).

*There are multiple issues with the figures - the use of icons is in many cases counterproductive and other representations are not clear or cause confusion (especially Figure 3).*

We have excluded former Fig. 3. Otherwise, the use of iconography is to facilitate graphical representation and cross-referencing between figures without over-cluttering. We provide all text and numeric values in the supporting information if individual detail is required.

*The phylogenetic information about the botanicals is missing. Also missing is any reference/discussion about how that analysis was able to disambiguate the confounding effects of shared uses and tastes of drugs from closely related species.*

This is explained in the methods (sections: ‘Phylogenetic tree’ and ‘statistical procedure’). We highlight that all models showed high heritability which means that shared ancestry has a statistical influence on the model. The trees themselves are now represented in our modified Figure 2.

#### **Reviewer #1 (Recommendations For The Authors):**

*Besides the points already covered in my public review, I believe it would be interesting to assess and discuss the differences between the category "food" (how many drugs were allocated there?) and the drugs used for therapeutic purposes. In this manner, the food category could serve as a retrospective negative control to test the authors' hypotheses. Does the food category include drugs of weak flavor? Does it include drugs of complex flavor?*

All drugs in this database are associated with therapeutic uses. Only 96 are specifically mentioned to be also used as food while in total at least 152 are also used as food (many of

the most obvious food drugs are not labelled as such in DMM). It is difficult to use the food category as a negative control (for testing whether food drugs have weaker tastes), because spices are included in the food category. If at all, only staples should be used for such an analysis. But this would be another study.

In the context of the present analyses, we do agree that there is interest and so we have therefore added a small section to our manuscript: The 96 botanical drugs specifically mentioned also for food (though there are more than 150 edible drugs in our dataset; Supplementary file 1) show positive associations with starchy (px = 0.005), nutty (px = 0.002) and salty (px = 0.001) and negative associations with bitter (px = 0.007), woody (px = 0.001) and stinging (px = 0.033) tastes and flavours.

*Please replace "plant defence" with "plant defense".*

Currently the whole MS is formatted BE. We are happy to revise on the basis of editorial policy.

**Reviewer #2 (Recommendations For The Authors):**

*1. I would encourage replacing "taste" with "flavor" throughout the manuscript and in the title because this paper addresses "taste here defined as a combination of taste, odour and chemesthesis" which essentially is the definition of flavor, and should not be simplified to taste. Flavor is the more precise word, and there is no need to confuse readers by defining "taste" in this way when taste means just the gustatory aspect of flavor.*

We now define flavour as a combination of taste, smell and chemesthesis and use 'taste' when referring to a specific taste quality. We use the term 'chemosensory' (perception, quality) and chemosensation for addressing the perception of both, taste and flavour qualities together. The abstract now reads: "The perception of taste and flavour (a combination of taste, smell and chemesthesis) here referred to as chemosensation, enables animals to find high-value foods and avoid toxins."

We prefer to leave the title as it is in accordance with standard books (e.g., "Pharmacology of Taste" by Palmer and Servant) which address all kinds of chemosensory interactions and the fact that we've conducted a 'tasting panel' (and not a 'flavour panel'), and because flavour as a concept is only used in English (and also there not consistently, with 'taste' being the preferred term used by English native speakers for describing perception where in a strict sense, 'flavour' would be the correct term, see Rozin P. "Taste-smell confusions" and the duality of the olfactory sense. Percept Psychophys. 1982 Apr;31(4):397-401)) and maybe also in French.

*1. Methods - A much more detailed description of how the samples were prepared for the taste tests is needed. Were they sampled as a dry powder? No, they were sampled as dried pieces. We have added more information to our methods section to clarify.*

Why is there such a big range in the amount provided (.1 to 2 g)? Because certain drugs are highly toxic (aconitum, opium) we could only provide a relatively small amount (that still permitted the perception of taste qualities). For practical reasons, half a walnut was dispensed. We have added more information to our methods section to clarify.

*Also "Panelists were instructed to spit, rinse their mouth with drinking water and to take a break before tasting the next sample" This seems more likely that the samples were*

*dissolved in a liquid if they were spitting and rinsing, but this is not clear. Also - take a break for how long between samples?*

Panelists were instructed to chew the amount of sample necessary for taste perception, to annotate their perception, and to spit out residues of samples and finally rinse their mouth with drinking water. The breaks between tasting different samples depended on chemosensory persistence. We have added more information to our methods section to clarify.

*How many samples were tested per day?*

The number of tasted samples was different from panelist to panelist and depending on available time frames. On average each panelist tasted 17,2 drugs per hour using 10.5 sessions (18 sessions in total) lasting approximately two hours each. We have added more information to our methods section to clarify.

*Did individual panelists get repeated samples?*

Random distribution permitted that individual panelists were challenged also with repeated samples. We have added more information to our methods section to clarify.

*1. Methods - Phylogenetic tree - Where is the output of this tree? It should be included in the figures and referred to in the results/discussion where the authors claim that they have been able to disambiguate phylogenetic closeness with taste and medicinal use.*

We did not 'build' a phylogenetic tree, rather we modified an existing one. Therefore, the wording of that section in the methods has been adjusted for clarity. We refer to the tree in the results pertaining to phylogenetic relatedness by explicitly quantifying the extent of phylogenetic signal using the widely used heritability ( $h^2$ ) statistic. This means that shared ancestry has a statistical influence on the model. We have also added to our Figure 2 representations of the phylogenetic tree we used in our analysis, limited to the species for which we have data, also displaying the data (in this case, intensity and complexity) at the tips.

*1. Taste intensity ratings should be better explained. Since the panelists are evaluating different amounts of samples (.1 to 2g) wouldn't the intensity of taste also depend on the amount of the substance?*

The panelists were not told to introduce all the sample into their mouth but just enough to perceive the taste qualities clearly (explanation given in methods). E.g.: one black pepper corn is normally enough to perceive the taste and flavour of pepper while the same amount of hazelnut would be insufficient.

*Or is this measure a relative value - "woodiness" vs "sourness" for example within the sample is strong/weak?*

Chemosensation and sensory perception in general is always relative. (For instance, currently I can hear the birds singing outside. Was there music playing in my room I wouldn't be able to hear them).

*Because of this - are samples with strong tastes less likely to seem complex because the intensity of one stimulus masks the other?*



Yes, we argue that drugs with strong tastes/flavours are less likely be perceived as being complex (fewer individual qualities perceived), arguably because strong stimuli overshadow weaker ones. We currently address this in the discussion and have made some modifications in line with the below comment.

*This issue was presented briefly in the discussion when addressing the finding that samples with intense, but fewer tastes were more versatile, but this was highly confusing.*

*The authors presented both sides of the problem without referring to any of their own experiments to resolve the issue, or to highlight this as a potential limitation of the study at hand.*

Yes, stronger tastes mask weaker tastes which addresses both sides of the problem.

We have modified the first paragraph of the discussion to make this clearer.

It now reads: "Unexpectedly, botanical drugs eliciting fewer but intense chemosensations were more versatile (Fig. 2). People often associate complexity with intensity, and taste complexity is popularly interpreted with a higher complexity of ingredients (Spence, and Wang, 2018). However, simple tastes can be associated with complex chemistry when intense tastes mask weaker tastes, or when tastants are blended (Breslin and Beauchamp, 1997; Green et al., 2010). For example, starchy flavours or sweet tastes can be sensed when bitter and astringent antifeedant compounds are present below a certain threshold while salts enhance overall flavour by suppressing the perception of bitter tastants (Breslin and Beauchamp, 1997; Johns, 1990). On the other hand, combinations of different tastants or olfactory stimuli do not necessarily result in increased perceived complexity (Spence and Wang, 2018; Weiss et al., 2012)."

*It would be useful to understand the parameters a bit more - a data visualization of the relationships of intensity and complexity across all samples would be a welcome addition to Figure 2.*

Shared ancestry has a statistical influence on the model. We have now also added to our Figure 2 representations of the phylogenetic tree we used in our analysis, limited to the species for which we have data, also displaying the data (in this case, intensity and complexity) at the tips.

*1. "Therapeutic Versatility" is a measure of how many different therapeutic uses a given botanic is listed in the DMM. This is one of the primary comparisons of this study, but the authors do not provide much of a rationale for using this metric. Also, there are 46 therapeutic uses, but many are interrelated such as gastric, gynecology, muscle, neurological, respiratory, skin, and kidney. It is not clear in my reading of the methods if this was also treated in some type of "phylogeny" as well or not. I would assume a real therapeutic versatility metric should be higher for something used for cough, ulcers, gout, and menses rather than something that was used for 4 different, but skin-related complaints.*

The reviewer is correct, and we appreciate this comment. We have modified the definition of versatility in line with the suggestions laid out here. We have provided a detailed explanation of this in our public responses but for ease of reference, we paste this again here:

Here, we define therapeutic versatility as the number of therapeutic 'categories' a drug is used for (the 25 broad categories are represented by shared iconography in Figure 1). Our



revised results include analyses using this definition – which are qualitatively identical to our previous results which defined versatility using the 46 individual therapeutic uses.

We repeated our original ‘versatility’ analyses using the 25 broader categories rather than the 46 individual uses. The results remained largely the same.

*1. Use of icons/pictorial representations in figures. Overall, the use of icons is not necessary - words could be used, and then readers would not need to keep going back and forth to the key in Figure 1 to identify the taste/use. I am very confused by Figure 3. How is the strength of taste shown in this figure? The use of the balance is a confusing representation since I don't associate strength/intensity with weight. Also there are specific tastes that are used more, and others that are used less (but the numbers of those are also more/less). I do not think this figure accomplishes the goal of relaying these findings.*

Whilst we agree that iconography is not strictly necessary, we think it is a good way of graphically representing the results without over-crowding the figures or introducing text sizes too small to read in print. All values are provided in the supporting information if any individual detail is required.

We have decided on the basis of these comments to exclude former Fig. 3 and (Figure supplement 1). We hope that the removal of this figure and clearer signposting towards the text and numerical tables in the supplementary information alleviates the reviewer's concerns.

*1. Similarly, figure 4 is unclear. This could be better represented in a table with words and p values listed. But a larger issue is that this shows essentially the same overarching relationship across the therapeutic use cases - high intensity, low complexity. Only the pink kidney (other?) case differs from this pattern. In the discussion, several therapeutic uses are discussed that could need intense tasting medicine - but these are not related directly back to the relationships shown in Figure 4.*

Yes, we agree with the reviewer and have now moved Fig. 4 to the supplementary (Figure supplement 1)